

Relatório Técnico

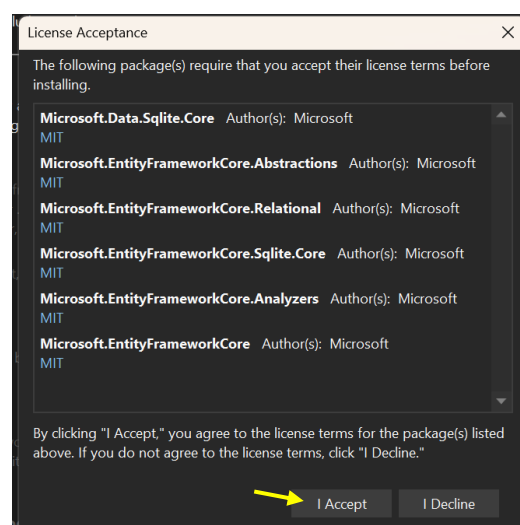
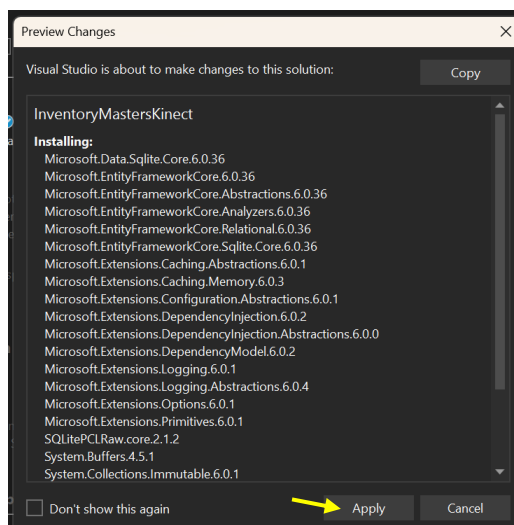
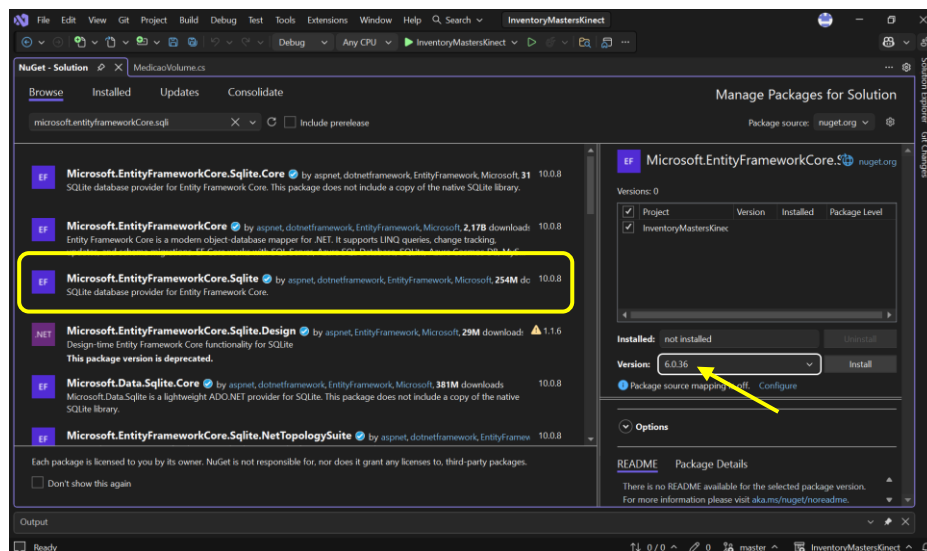
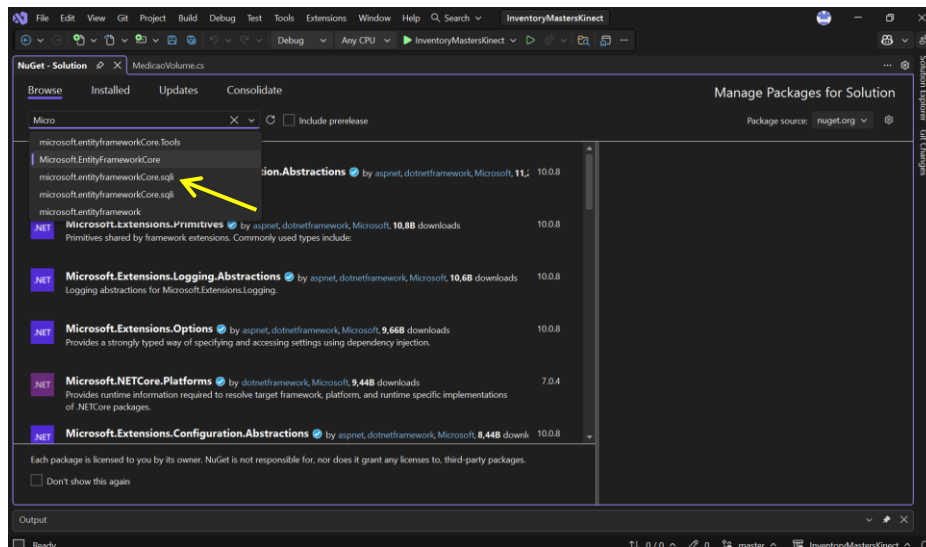
Configuração e Inicialização do SQLite no EF Core (.NET 6)

Passo 1: Gerenciamento de Pacotes NuGet (Instalações)

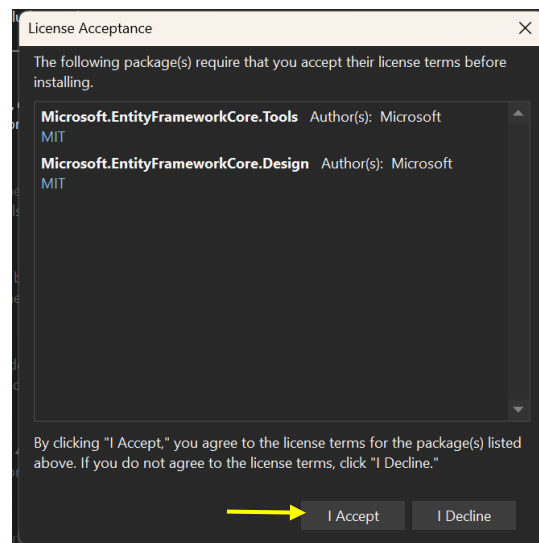
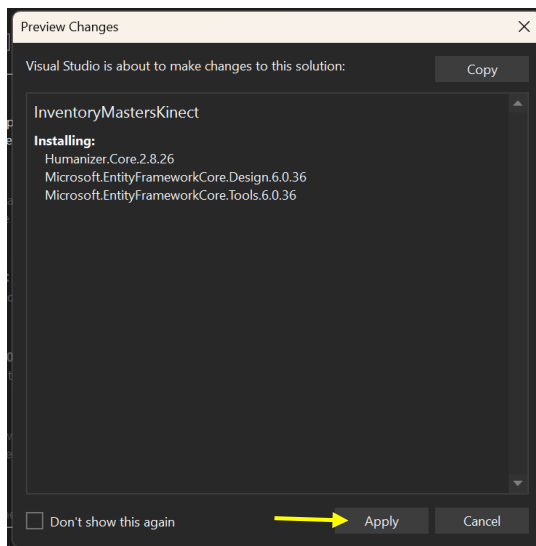
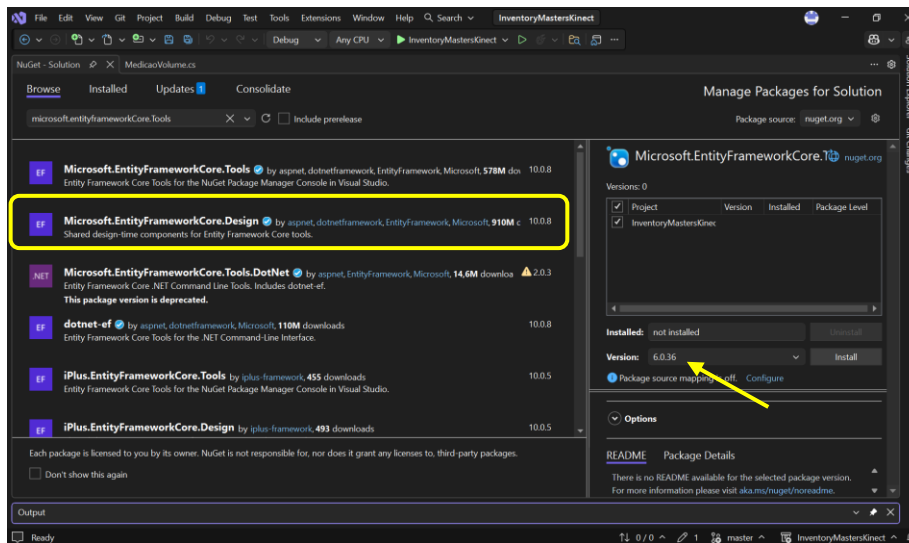
- No menu superior do Visual Studio 2022, acesse: Tools (Ferramentas) > NuGet Package Manager > Manage NuGet Packages for Solution....
- Vá na aba **Browse** (Procurar) e pesquise por Microsoft.EntityFrameworkCore.Sqlite.
- Selecione o projeto InventoryMastersKinect, mude a versão fixada para **6.0.36** (versão LTS totalmente compatível com o .NET 6) e clique em **Install**.
- Repita o mesmo procedimento exato de versão para os pacotes:
- Microsoft.EntityFrameworkCore.Design (6.0.36)
- Microsoft.EntityFrameworkCore.Tools (6.0.36)
- Adicione também o pacote SQLitePCLRaw.bundle_e_sqlite3 na versão **2.1.8** para garantir a inicialização correta das dependências nativas de C++ do SQLite.

1.1 – MicrosoftEntityFrameworkCore.Sqli

TCC – Inventory Masters
Alunos: Danilo, Diulie, Marilene e Miguel
Curso: Desenvolvimento de Sistemas
Turno: Noite



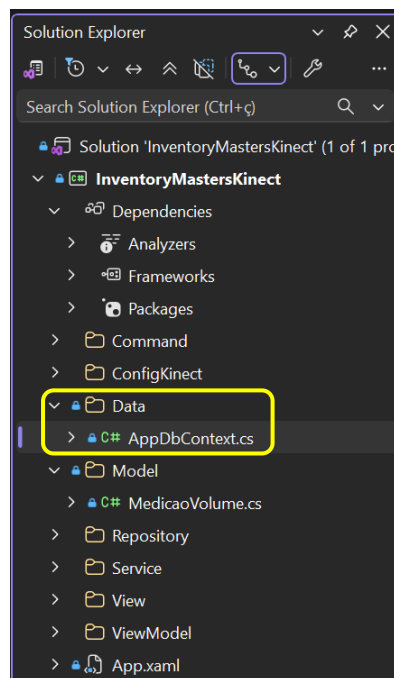
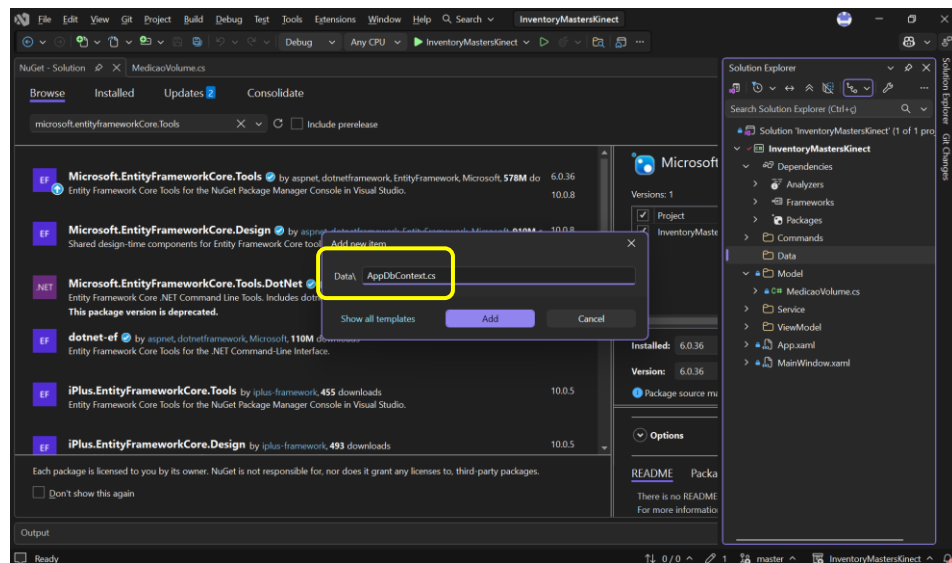
1.2 – MicrosoftEntityFrameworkCore.Tools



Passo 2: Criação da Estrutura de Código (AppDbContext)

- No *Solution Explorer*, clique com o botão direito no projeto ➡ Add ➡ New Folder e nomeie como Data.
- Dentro da pasta, adicione uma nova classe chamada AppDbContext.cs.
- Declare a coleção que representará sua tabela (o mapeamento das leituras do sensor Kinect): `public DbSet<MedicaoVolume> MedicoesVolume { get; set; }.`

- Sobrescreva o método OnConfiguring para injetar a string de conexão local isolada: `optionsBuilder.UseSqlite("Data Source=inventory_masters.db");`.



Passo 3: Abra o Console do Gerenciador de Pacotes (Package Manager Console)

- No menu superior, acesse: Tools > NuGet Package Manager > Package Manager Console.

- **Atenção:** Mude o seletor suspenso **Default project** (Projeto padrão) no topo do console para refletir InventoryMastersKinect.
- Execute o comando de criação do mapeamento relacional:

```
PowerShell  
PM> Add-Migration InitialCreate
```

- Uma pasta física chamada Migrations/ nascerá na árvore do seu projeto, contendo os arquivos em C# que traduzem seus modelos para tabelas estruturadas.

Passo 4: A Criação Física do Banco (Update-Database)

- Com o histórico gerado, execute no terminal do console o comando de aplicação definitiva:

```
PowerShell  
PM> Update-Database
```

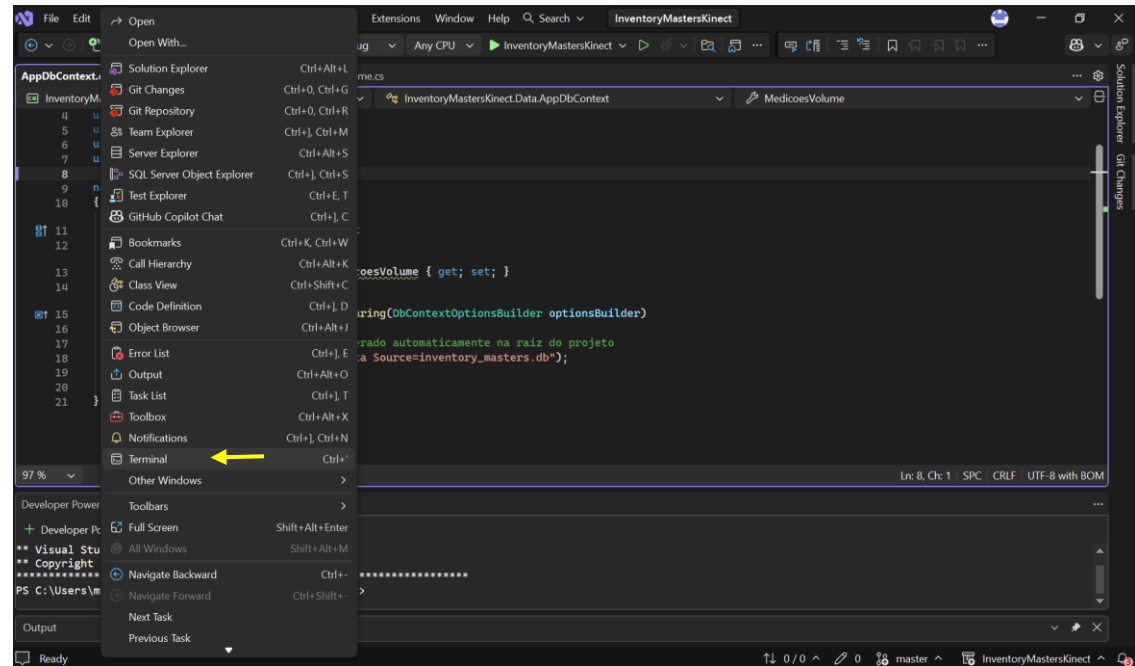
- O Entity Framework Core lerá os scripts e criará automaticamente o arquivo autônomo **inventory_masters.db** diretamente na pasta do diretório de execução do software.
- A partir deste ponto, o ambiente local de desenvolvimento está completamente integrado e pronto para efetuar operações de persistência de dados.

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```
PS C:\Users\maryl\source\repos\InventoryMastersKinect> cd InventoryMastersKinect
PS C:\Users\maryl\source\repos\InventoryMastersKinect\InventoryMastersKinect> dotnet ef migrations add InitialCreate
```

```
Developer PowerShell
+ Developer PowerShell
PS C:\Users\maryl\source\repos\InventoryMastersKinect> cd InventoryMastersKinect
PS C:\Users\maryl\source\repos\InventoryMastersKinect\InventoryMastersKinect> dotnet ef migrations add InitialCreate
Build started...
Build succeeded.
You must install or update .NET to run this application.

App: C:\Users\maryl\.dotnet\tools\store\dotnet-ef\10.0.5\dotnet-ef\10.0.5\tools\net8.0\any\tools\net8.0\any\ef.dll
Architecture: x64
Framework: 'Microsoft.NETCore.App', version '6.0.0' (x64)
.NET location: C:\Program Files\dotnet\

The following frameworks were found:
 8.0.24 at [C:\Program Files\dotnet\shared\Microsoft.NETCore.App]
 8.0.27 at [C:\Program Files\dotnet\shared\Microsoft.NETCore.App]
10.0.8 at [C:\Program Files\dotnet\shared\Microsoft.NETCore.App]

The following frameworks for other architectures were found:
x86
 8.0.27 at [C:\Program Files (x86)\dotnet\shared\Microsoft.NETCore.App]
```

```
Developer PowerShell
+ Developer PowerShell
PS C:\Users\maryl\source\repos\InventoryMastersKinect\InventoryMastersKinect> dotnet ef database update
Build started...
Build succeeded.
You must install or update .NET to run this application.

App: C:\Users\maryl\.dotnet\tools\store\dotnet-ef\10.0.5\dotnet-ef\10.0.5\tools\net8.0\any\tools\net8.0\any\ef.dll
Architecture: x64
Framework: 'Microsoft.NETCore.App', version '6.0.0' (x64)
.NET location: C:\Program Files\dotnet\

The following frameworks were found:
 8.0.24 at [C:\Program Files\dotnet\shared\Microsoft.NETCore.App]
 8.0.27 at [C:\Program Files\dotnet\shared\Microsoft.NETCore.App]
10.0.8 at [C:\Program Files\dotnet\shared\Microsoft.NETCore.App]

The following frameworks for other architectures were found:
x86
 8.0.27 at [C:\Program Files (x86)\dotnet\shared\Microsoft.NETCore.App]
10.0.8 at [C:\Program Files (x86)\dotnet\shared\Microsoft.NETCore.App]
```

```
PS C:\Users\maryl\source\repos\InventoryMastersKinect\InventoryMastersKinect> winget install Microsoft.DotNet.Runtime.6
The 'msstore' source requires that you view the following agreements before using.
Terms of Transaction: https://aka.ms/microsoft-store-terms-of-transaction
The source requires the current machine's 2-letter geographic region to be sent to the backend service to function properly (ex. "US").
Do you agree to all the source agreements terms?
[Y] Yes [N] No: Y
```

```
Do you agree to all the source agreements terms?
[Y] Yes [N] No: Y
Found Microsoft .NET Runtime 6.0 [Microsoft.DotNet.Runtime.6] Version 6.0.36
This application is licensed to you by its owner.
Microsoft is not responsible for, nor does it grant any licenses to, third-party packages.
Downloading https://builds.dotnet.microsoft.com/dotnet/Runtime/6.0.36/dotnet-runtime-6.0.36-win-x64.exe
26.7 MB / 26.7 MB
Successfully verified installer hash
Starting package install...
Successfully installed
```

Passo 5: Reiniciar o Visual Studio e executar os comandos abaixo:

```
Developer PowerShell
+ Developer PowerShell
PS C:\Users\maryl\source\repos\InventoryMastersKinect> cd InventoryMastersKinect
PS C:\Users\maryl\source\repos\InventoryMastersKinect\InventoryMastersKinect> dotnet ef migrations add InitialCreate
Build started...
Build succeeded.
You must install or update .NET to run this application.

App: C:\Users\maryl\.dotnet\tools\.store\dotnet-ef\10.0.5\dotnet-ef\10.0.5\tools\net8.0\any\tools\net8.0\any\ef.dll
Architecture: x64
Framework: 'Microsoft.WindowsDesktop.App', version '6.0.0' (x64)
.NET location: C:\Program Files\dotnet\

The following frameworks were found:
 8.0.24 at [C:\Program Files\dotnet\shared\Microsoft.WindowsDesktop.App]
 8.0.27 at [C:\Program Files\dotnet\shared\Microsoft.WindowsDesktop.App]
10.0.8 at [C:\Program Files\dotnet\shared\Microsoft.WindowsDesktop.App]

The following frameworks for other architectures were found:
 x86
 8.0.27 at [C:\Program Files (x86)\dotnet\shared\Microsoft.WindowsDesktop.App]

Developer PowerShell
+ Developer PowerShell
Framework: 'Microsoft.WindowsDesktop.App', version '6.0.0' (x64)
.NET location: C:\Program Files\dotnet\

The following frameworks were found:
 8.0.24 at [C:\Program Files\dotnet\shared\Microsoft.WindowsDesktop.App]
 8.0.27 at [C:\Program Files\dotnet\shared\Microsoft.WindowsDesktop.App]
10.0.8 at [C:\Program Files\dotnet\shared\Microsoft.WindowsDesktop.App]

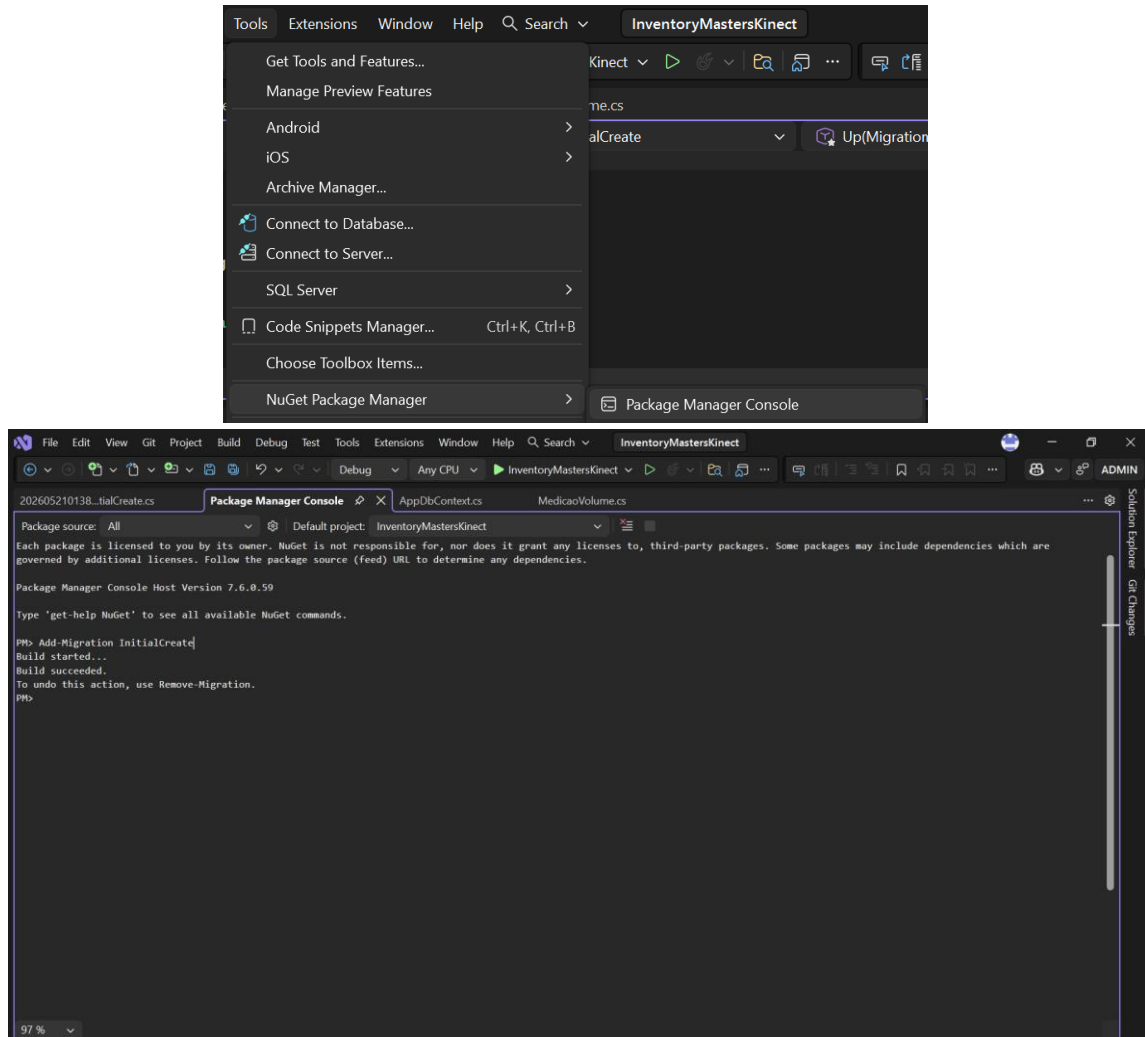
The following frameworks for other architectures were found:
 x86
 8.0.27 at [C:\Program Files (x86)\dotnet\shared\Microsoft.WindowsDesktop.App]
10.0.8 at [C:\Program Files (x86)\dotnet\shared\Microsoft.WindowsDesktop.App]

Learn more:
https://aka.ms/dotnet/app-launch-failed

To install missing framework, download:
https://aka.ms/dotnet-core-applaunch?framework=Microsoft.WindowsDesktop.App&framework_version=6.0.0&arch=x64&rid=win-x64&os=win10
PS C:\Users\maryl\source\repos\InventoryMastersKinect\InventoryMastersKinect>
PS C:\Users\maryl\source\repos\InventoryMastersKinect\InventoryMastersKinect> winget install Microsoft.DotNet.DesktopRuntime.6
Found Microsoft .NET Windows Desktop Runtime 6.0 [Microsoft.DotNet.DesktopRuntime.6] Version 6.0.36
This application is licensed to you by its owner.
Microsoft is not responsible for, nor does it grant any licenses to, third-party packages.
Downloading https://builds.dotnet.microsoft.com/dotnet/WindowsDesktop/6.0.36/windowsdesktop-runtime-6.0.36-win-x64.exe
54.7 MB / 54.7 MB
Successfully verified installer hash
Starting package install...
Successfully installed

Developer PowerShell
+ Developer PowerShell
** Visual Studio 2026 Developer PowerShell v18.6.1
** Copyright (c) 2026 Microsoft Corporation
*****
PS C:\Users\maryl\source\repos\InventoryMastersKinect> dotnet ef migrations add InitialCreate
No project was found. Change the current working directory or use the --project option.
PS C:\Users\maryl\source\repos\InventoryMastersKinect> cd InventoryMastersKinect
PS C:\Users\maryl\source\repos\InventoryMastersKinect\InventoryMastersKinect> dotnet ef migrations add InitialCreate
Build started...
Build succeeded.
Unhandled exception: System.IO.FileLoadException: Could not load file or assembly 'System.Runtime, Version=8.0.0.0, Culture=neutral, PublicKeyToken=b03f5f7f11d50a3a'.
The located assembly's manifest definition does not match the assembly reference. (0x80131040)
File name: 'System.Runtime, Version=8.0.0.0, Culture=neutral, PublicKeyToken=b03f5f7f11d50a3a'
PS C:\Users\maryl\source\repos\InventoryMastersKinect\InventoryMastersKinect>
```

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```
202605210138...tialCreate.cs Package Manager Console AppDbContext.cs MedicaoVolume.cs
Package source: All Default project: InventoryMastersKinect
Each package is licensed to you by its owner. NuGet is not responsible for, nor does it grant any licenses to, third-party packages. Some packages may include dependencies which are governed by additional licenses. Follow the package source (feed) URL to determine any dependencies.
Package Manager Console Host Version 7.6.0.59
Type 'get-help NuGet' to see all available NuGet commands.
PM> Add-Migration InitialCreate
Build started...
Build succeeded.
To undo this action, use Remove-Migration.
PM> Update-Database
Build started...
Build succeeded.
System.Exception: You need to call SQLitePCL.raw.SetProvider(). If you are using a bundle package, this is done by calling SQLitePCL.Batteries.Init().
    at SQLitePCL.raw.get_Provider()
    at SQLitePCL.raw.sqlite3_open_v2(utf8z filename, sqlite3* db, Int32 flags, utf8z vfs)
    at SQLitePCL.raw.sqlite3_open_v2(String filename, sqlite3* db, Int32 flags, String vfs)
    at Microsoft.Data.Sqlite.SQLiteConnectionInternal..ctor(SQLiteConnectionStringBuilder connectionOptions, SQLiteConnectionPool pool)
    at Microsoft.Data.Sqlite.SQLiteConnectionFactory.GetConnection(SQLiteConnection outerConnection)
    at Microsoft.Data.Sqlite.SQLiteConnection.Open()
    at Microsoft.EntityFrameworkCore.Storage.RelationalConnection.OpenDbConnection(Boolean errorsExpected)
    at Microsoft.EntityFrameworkCore.Storage.RelationalConnection.OpenInternal(Boolean errorsExpected)
    at Microsoft.EntityFrameworkCore.Storage.RelationalConnection.Open(Boolean errorsExpected)
    at Microsoft.EntityFrameworkCore.Sqlite.Storage.Internal.SqliteDatabaseCreator.Exists()
    at Microsoft.EntityFrameworkCore.Migrations.HistoryRepository.Exists()
    at Microsoft.EntityFrameworkCore.Migrations.Internal.Migrator.Migrate(String targetMigration)
    at Microsoft.EntityFrameworkCore.Design.Internal.MigrationsOperations.UpdateDatabase(String targetMigration, String connectionString, String contextType)
    at Microsoft.EntityFrameworkCore.Design.OperationExecutor.UpdateDatabaseImpl(String targetMigration, String connectionString, String contextType)
    at Microsoft.EntityFrameworkCore.Design.OperationExecutor.<.>c__DisplayClass0_0.<.ctor>b__0()
    at Microsoft.EntityFrameworkCore.Design.OperationExecutor.OperationBase.Execute(Action action)
You need to call SQLitePCL.raw.SetProvider(). If you are using a bundle package, this is done by calling SQLitePCL.Batteries.Init().
PM> |
97 %
PM> Install-Package SQLitePCL.bundle_e_sqlite3 -Version 2.1.8
Restoring packages for C:\Users\maryl\source\repos\InventoryMastersKinect\InventoryMastersKinect\InventoryMastersKinect.csproj...
GET https://api.nuget.org/v3-flatcontainer/sqlitepclraw.bundle_e_sqlite3/index.json
OK https://api.nuget.org/v3-flatcontainer/sqlitepclraw.bundle_e_sqlite3/index.json 471ms
GET https://api.nuget.org/v3-flatcontainer/sqlitepclraw.bundle_e_sqlite3/2.1.8/sqlitepclraw.bundle_e_sqlite3.2.1.8.nupkg
OK https://api.nuget.org/v3-flatcontainer/sqlitepclraw.bundle_e_sqlite3/2.1.8/sqlitepclraw.bundle_e_sqlite3.2.1.8.nupkg 199ms
GET https://api.nuget.org/v3-flatcontainer/sqlitepclraw.provider_e_sqlite3/index.json
GET https://api.nuget.org/v3-flatcontainer/sqlitepclraw.lib_e_sqlite3/index.json
OK https://api.nuget.org/v3-flatcontainer/sqlitepclraw.provider_e_sqlite3/index.json 217ms
GET https://api.nuget.org/v3-flatcontainer/sqlitepclraw.provider_e_sqlite3/2.1.8/sqlitepclraw.provider_e_sqlite3.2.1.8.nupkg
OK https://api.nuget.org/v3-flatcontainer/sqlitepclraw.provider_e_sqlite3/2.1.8/sqlitepclraw.provider_e_sqlite3.2.1.8.nupkg 99ms
GET https://api.nuget.org/v3-flatcontainer/sqlitepclraw.lib_e_sqlite3/index.json
GET https://api.nuget.org/v3-flatcontainer/sqlitepclraw.lib_e_sqlite3/2.1.8/sqlitepclraw.lib_e_sqlite3.2.1.8.nupkg
OK https://api.nuget.org/v3-flatcontainer/sqlitepclraw.lib_e_sqlite3/2.1.8/sqlitepclraw.lib_e_sqlite3.2.1.8.nupkg 76ms
GET https://api.nuget.org/v3-flatcontainer/sqlitepclraw.core/index.json
OK https://api.nuget.org/v3-flatcontainer/sqlitepclraw.core/index.json 292ms
GET https://api.nuget.org/v3-flatcontainer/sqlitepclraw.core/2.1.8/sqlitepclraw.core.2.1.8.nupkg
OK https://api.nuget.org/v3-flatcontainer/sqlitepclraw.core/2.1.8/sqlitepclraw.core.2.1.8.nupkg 120ms
97 %
```

```
PM> Update-Database
Build started...
Build succeeded.
Applying migration '20260521013823_InitialCreate'.
Done.
PM>
```

Anexo I

O Checklist para a Equipe

1. O SDK do .NET 6 e os Runtimes Corretos

Os colaboradores precisam garantir que o ambiente do .NET 6 está completo para aplicações Desktop (WPF). Se eles já utilizam o Visual Studio com a carga de trabalho para desenvolvimento desktop instalada, a maioria dos runtimes já deve estar presente. Caso contrário, o comando rápido via terminal resolve:

PowerShell

`winget install Microsoft.DotNet.DesktopRuntime.6`

2. A Ferramenta Global do Entity Framework (dotnet-ef)

Se algum colaborador precisar criar novas tabelas ou rodar comandos de *migration* futuramente, ele precisará da ferramenta de linha de comando do EF Core. Instale a versão correspondente à do projeto executando:

Bash

`dotnet tool install --global dotnet-ef --version 6.0.36`

3. O Driver Nativo do SQLite (se necessário no SO)

Como a estrutura base do banco já foi criada, o pacote SQLitePCLRaw.bundle_e_sqlite3 já foi integrado ao projeto e enviado para o Git através dos *commits*. Portanto, as dependências do SQLite serão restauradas

automaticamente na máquina dos demais membros assim que eles executarem um dotnet restore ou compilarem o projeto pela primeira vez.

O que a equipe precisa fazer ao puxar o código (*Git Pull*)?

Assim que os outros desenvolvedores derem git pull e pegarem a branch atualizada, eles só precisam seguir estes passos rápidos para criar o banco de dados local em suas respectivas máquinas:

1. Abrir o **Package Manager Console** no Visual Studio (Tools > NuGet Package Manager > Package Manager Console).
2. Garantir que o projeto padrão (**Default project**) selecionado no topo do console é o **InventoryMastersKinect**.
3. Rodar o comando de atualização para gerar o arquivo do banco de dados:

PowerShell

Update-Database